

VelaDial Theme Visual Upgrade Plan

Theme Visual Upgrade Plan

VelaDial Theme Visual Upgrade Plan

Status: This plan documents the visual upgrade path for the multi-theme firmware candidate. PR #56 creates the integrated 20-theme firmware engine foundation. Further visual refinement may be needed after hardware testing and contact-sheet review. Current goal is one firmware with all 20 selectable themes, not final visual perfection.

1. Critique of Current Concept Prototypes

The initial 20 concept prototypes served their purpose: proving that 20 distinct metaphors could be compiled and rendered within ESPHome's LVGL constraints. However, as noted in the visual contact sheet review, they currently look too "AI-generated" and basic.

Why they look basic:

- **Over-reliance on primitive geometry:** Too many themes rely on a single centered circle or a basic arc with default thickness.
- **Generic typography:** The default Roboto font is used everywhere without varying weight, tracking, or size to create hierarchy.
- **Lack of depth:** Most themes are flat, using solid colors without layering, opacity blending, or subtle borders to create a premium feel.
- **Literal interpretations:** Some metaphors (like the door switch or Apple Watch complications) are too literal and lack the abstraction expected in a luxury smart home device.

2. The Upgrade Strategy

To elevate these 20 themes into a premium, production-ready firmware, we will apply a unified design system that upgrades each theme while maintaining its unique identity.

A. Stronger Visual Hierarchy

- **Typography:** We will use `roboto48` for primary data (brightness percentage), `roboto24` for secondary data (power state, preset names), and `roboto16` for tertiary labels.
- **Contrast:** We will ensure high contrast for active elements (e.g., `0xFFB000` amber) and low contrast for inactive elements (e.g., `0x333333` dark gray).

B. Refined Spacing and Composition

- **Breathing room:** We will move away from the "generic circle + text" layout by utilizing the full 240×240 canvas. Elements will be pushed toward the edges (e.g., outer arcs) or clustered deliberately to create negative space.
- **Alignment:** All text and icons will be perfectly centered or aligned to specific grids, avoiding the "floating" look of early prototypes.

C. Richer Color Palettes

- **Thematic colors:** While amber (`0xFFB000`) remains the primary active color, themes will use specific palettes (e.g., deep blues for Tidal Wave, warm browns for Tree Ring, stark white/black for Minimal).
- **Opacity blending:** We will use `bg_opa` to create subtle glows and layered effects, especially in themes like Eclipse Corona and Flame Flicker.

D. Polished LED Ring Behavior

- The physical WS2812 LED ring will no longer just turn on/off. It will mirror the active theme's primary color and intensity, extending the UI beyond the screen onto the wall.

3. Themes Requiring the Most Improvement

1. **Theme 03 (Large Center Power):** Needs to move from a basic green circle to a refined, glowing power emblem.
2. **Theme 08 (Apple Watch Complications):** Needs to look less like a smartwatch clone and more like a sophisticated data dashboard.
3. **Theme 12 (Door Switch Replacement):** Needs to abstract the physical switch into a sleek, digital toggle rather than a literal drawing.

4. The Theme Selector UX

The Theme Selector itself must feel premium. It will not be a basic list. - **Interaction:** Long-press the knob to enter. Rotate to scroll. Press to apply. - **Visuals:** It will display the theme number (e.g., "13/20"), the theme name in bold, and a subtle instruction ("Press to apply"). - **Persistence:** The selection will be saved to flash memory (`restore_value: yes`) so the device remembers its identity across reboots.